

Polynomial Regression: Geothermal Plant Modelling

Roll no.: BT2024095 Code (GitHub): [INSERT YOUR REPO LINK HERE]

1. Problem and data

Two datasets are modelled with polynomial regression only. **var1** predicts the Net Power Score of a steam turbine from six operating parameters (stated degree up to 10). **var2** predicts the Thermal Anomaly Score at 3-D coordinates (stated degree up to 20). Each has 1000 training rows and 1000 test rows, with no missing values. The metrics are MSE and R^2 .

	var1 (turbine)	var2 (thermal)
Features / train rows / test rows	6 / 1000 / 1000	3 / 1000 / 1000
y mean	0.93	2.61
y std	3.26	6.95
y min	-10.03	-22.34
y max	14.89	39.30
Feature range	[-1, 1]	[-1, 1]
Inputs exactly at ± 1	32%	24%

Table 1. Dataset summary. A large share of inputs sit exactly at ± 1 , so the generator clipped them to [-1, 1].

2. Method

Features. Every monomial whose total degree (sum of powers) is at most d is generated, as the assignment defines degree. The number of terms (without the constant) is $C(n+d, d) - 1$, which grows fast (Table 2). All columns are standardised using training-fold statistics only.

degree d	3	4	5	6	7	8	10	11	14
var1 (n=6)	83	209	461	923	1715	3002	8007	12375	38759
var2 (n=3)	19	34	55	83	119	164	285	363	679

Table 2. Number of polynomial terms. With 1000 samples (800 per CV fold), var1 at degree 6+ and var2 at degree 14+ approach or exceed the sample count.

Three fitting methods were compared on identical features: **OLS** (no penalty), **Ridge** (L2 penalty) and **Lasso** (L1 penalty, which also drops terms). All are linear in the polynomial coefficients, so this is still polynomial regression.

Selection. Degree and penalty strength α were chosen jointly by **5-fold cross-validation** (shuffled, seed 0) on the training data, minimising MSE. The grid was degree 1-7 for var1 and 2-14 for var2, with 4-5 α values per penalised method. An independent 80/20 holdout split was then used once to sanity-check the chosen setting. The final model is refit on all 1000 rows.

3. Results

Degree sweep. Table 3 gives the best CV MSE for each degree and method (alpha tuned per cell). The OLS columns show why regularisation is needed.

deg	var1 OLS	var1 Ridge	var1 Lasso		var2 OLS	var2 Ridge	var2 Lasso
1	9.209	9.206	9.209		-	-	-
2	3.099	3.096	3.085		21.040	21.037	21.040
3	1.042	1.041	1.008		13.290	13.288	13.276
4	0.853	0.783	0.642		3.762	3.762	3.763
5	2.190	0.578	0.355		1.677	1.677	1.677
6	158	0.688	0.377		0.589	0.588	0.590
7	10.773	0.709	0.384		0.348	0.348	0.363
8	-	-	-		0.274	0.271	0.293
9	-	-	-		0.308	0.265	0.278
10	-	-	-		0.399	0.263	0.268
11	-	-	-		0.496	0.258	0.267
12	-	-	-		1.815	0.261	0.267
13	-	-	-		6.693	0.264	0.264
14	-	-	-		198	0.270	0.267

Table 3. 5-fold CV MSE by degree (lower is better; highlighted rows = chosen degrees: 5 for var1, 11 for var2; a dash means degree not searched). OLS degrades sharply once terms approach the sample size.

	Method	Best degree	Best alpha	CV MSE	CV R2
var1	OLS	4	-	0.853	0.9194
var1	Ridge	5	10	0.578	0.9454
var1	Lasso	5	0.008	0.355	0.9665
var2	OLS	8	-	0.274	0.9943
var2	Ridge	11	1	0.258	0.9946
var2	Lasso	13	0.001	0.264	0.9945

Table 4. Best setting found for each method over the whole grid.

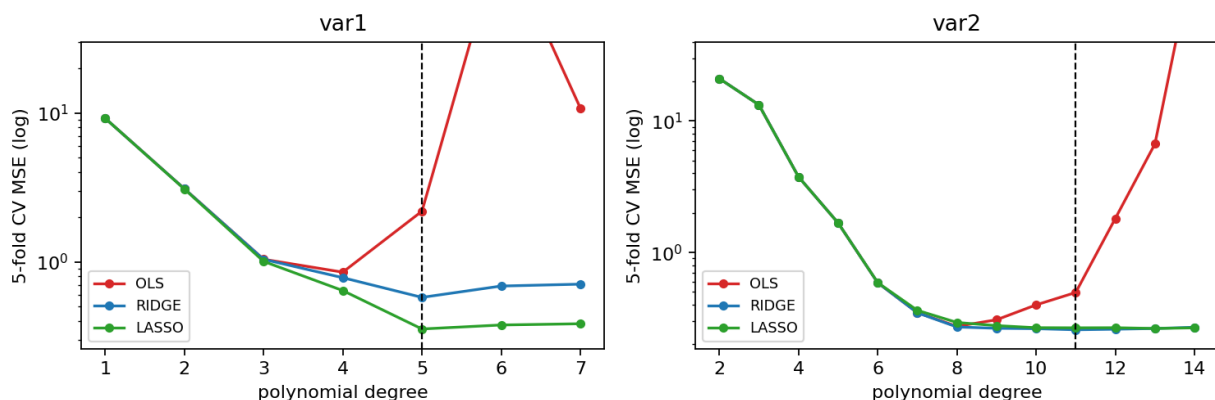


Figure 1. CV MSE vs degree for each method (log scale; dashed line = chosen degree).

3.1 var1: turbine net power

Degrees 1-3 underfit ($MSE \geq 1.0$). OLS is best at degree 4 (MSE 0.853) and then collapses (158 at degree 6 with 924 terms). Ridge improves to 0.578 at degree 5 (α 10). **Lasso is clearly best: 0.355 at degree 5**, a 39% reduction over Ridge. The best Lasso fit keeps only a fraction of terms, which indicates the true surface is sparse (a few interaction terms dominate). Degrees 6-7 with Lasso are slightly worse (0.377, 0.384), so degree 5 is the choice. The top terms are in Table 5.

rank	term	std. coef
1	$x_1 \cdot x_5$	-1.383
2	$x_2^3 \cdot x_3$	-0.986
3	$x_3 \cdot x_6$	-0.866
4	x_6^2	+0.782
5	$x_3^3 \cdot x_6^2$	-0.693
6	$x_2 \cdot x_4 \cdot x_6$	-0.596
7	$x_2 \cdot x_6$	+0.526
8	$x_2^2 \cdot x_4^2 \cdot x_5$	-0.504

Table 5. Largest standardised coefficients, final var1 Lasso (139 of 461 terms non-zero).

3.2 var2: thermal anomaly

Error falls steeply until degree 7-8 and then flattens near 0.26. Here the surface is dense: OLS at degree 8 already reaches 0.274 and Lasso does not beat Ridge. Ridge gives the lowest MSE, **0.258 at degree 11 (alpha 1)**. Its curve is flat across degrees 9-13 (0.258-0.265), so the exact degree is not critical. Without a penalty the fit blows up at high degree (OLS 1.82 at degree 12 and 198 at degree 14), while Ridge stays at 0.270 at degree 14. That plateau is consistent with an irreducible noise level of roughly 0.25 in y , which no polynomial can remove. The brief mentions degree up to 20, but 1000 points do not support such a model and CV shows no gain beyond degree 11.

3.3 Final models

	Method	Degree	alpha	Terms (non-zero)	CV MSE	CV R2	Holdout MSE	Holdout R2
var1	Lasso	5	0.008	461 (139)	0.355	0.9665	0.301	0.9763
var2	Ridge	11	1	363 (363)	0.258	0.9946	0.260	0.9936

Table 6. Final configuration and scores. Test-set scores are unavailable (hidden labels); CV and holdout are the estimates.

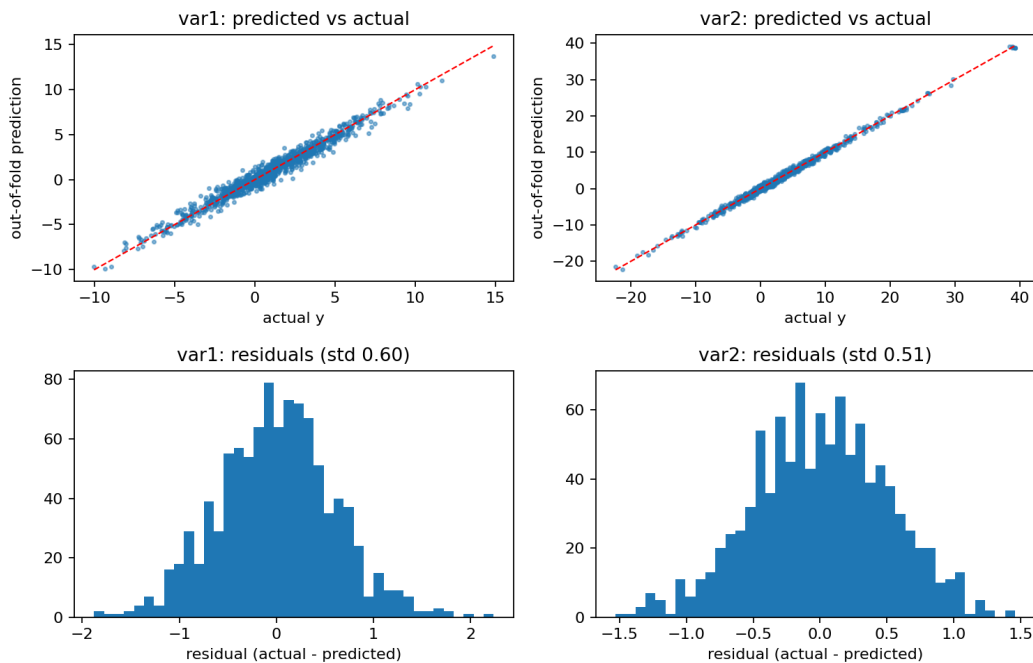


Figure 2. Out-of-fold predictions (top) and residuals (bottom) for the final models. Points hug the diagonal and residuals are centred and roughly symmetric, with no obvious structure left.

4. Discussion

Why regularise. Monomials on $[-1, 1]$ are strongly correlated, and the term count is comparable to the sample size, so plain least squares has high variance at higher degrees. Ridge shrinks all coefficients and is best when many terms contribute (var2). Lasso zeroes out terms and is best when few do (var1). Using CV to choose the degree *and* penalty together avoids picking degree by training error.

Limitations. (i) Degree and alpha are chosen on the same CV folds that report the score, so the CV MSE is slightly optimistic; the separate holdout agrees (var1 0.30, var2 0.26) but is based on only 200 points and is noisy. (ii) Lasso uses a loose solver tolerance ($1e-3$) for speed. (iii) The noise-level interpretation for var2 is an inference from the plateau, not a measured quantity. (iv) Test performance may differ if the test points cover regions the training sample covers poorly, particularly the clipped edges at ± 1 .

5. Reproducibility

Code file: `train.py` (degree/alpha search, final fit, test predictions). Run `python train.py` with the data in `data/`. Outputs: `BT2024095_pred_var1.csv` and `BT2024095_pred_var2.csv` (one column y, 1000 rows each).